Project Design Phase
Solution Architecture

Date	11th October 2026
Team ID	ServiceNow - A2
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

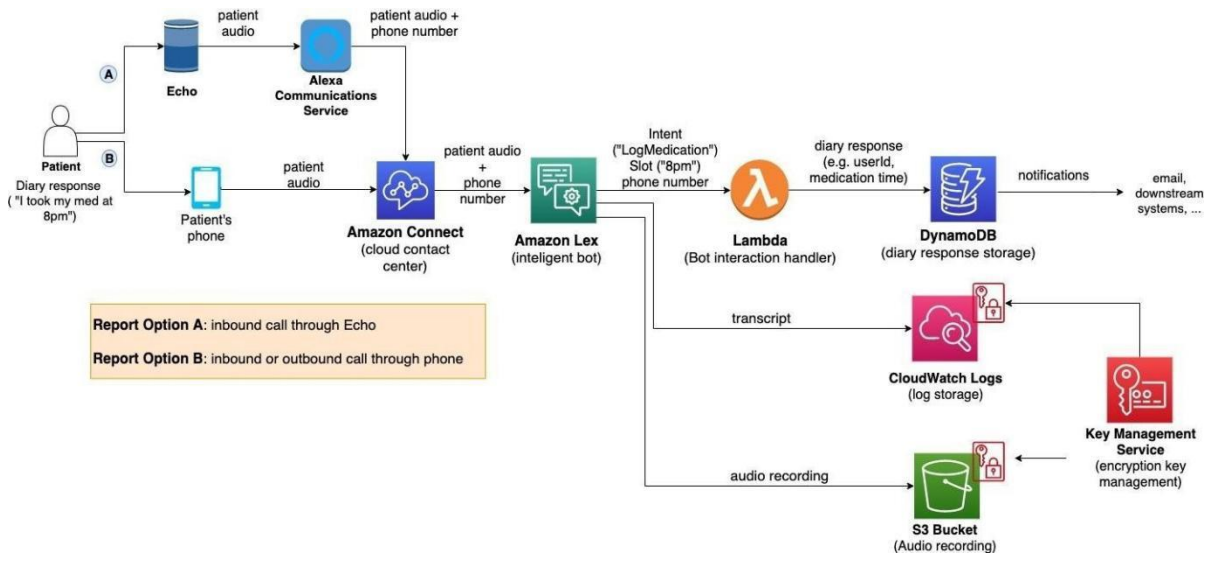


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Solution Architecture – Car Rental Request Automation

The architecture is centered on ServiceNow. The Service Catalog provides the employee entry point, while Request/RITM records, Flow Designer/Workflow, approvals, assignment groups and notifications coordinate the fulfillment lifecycle.

Logical Architecture

Employee / Requester
↓ ServiceNow Service Catalog – Car Rental Request
↓ Request + RITM Creation / Validation
↓ Flow Designer / Workflow
↓ Approval → Transport Assignment
↓ Availability Check → Fulfillment → Notifications
↓ Request Completed / Tracked

Architecture Components

Layer	ServiceNow Component	Responsibility	Output
Presentation	Service Catalog / Catalog Item	Collect employee request details.	Submitted catalog request
Request Data	Request + RITM + Variables	Store request and variable values.	Trackable RITM
Automation	Flow Designer / Workflow	Control sequence and conditions.	Automated process
Governance	Approval	Obtain authorization.	Approval decision
Fulfillment	Assignment Group / Transport Team	Review availability and arrange vehicle.	Fulfillment details
Communication	Notifications	Inform requester and teams.	Status updates
Security	Roles / ACLs	Restrict access to appropriate users.	Controlled access

Design Principles

- Keep the catalog form simple and focused.
- Capture required information at the point of submission.
- Automate repetitive routing and communication.
- Keep approval and fulfillment traceable through the RITM.
- Apply role-based access and maintain an audit trail.